

- RAG Learning

Topic 15 — Forces and Matter

Edexcel GCSE (9-1) Physics (1PH0) — Paper 2

15.1 Elastic Behaviour and Hooke's Law

$$F = k \times x$$

Valid only up to the **limit of proportionality**. Up to the **elastic limit**, deformation is elastic (object returns to original shape); beyond it, deformation is inelastic/plastic (permanent).

$$\text{Elastic PE} = \frac{1}{2} k x^2$$

Worked Example: Spring of $k = 250 \text{ N/m}$ stretched by 0.10 m .

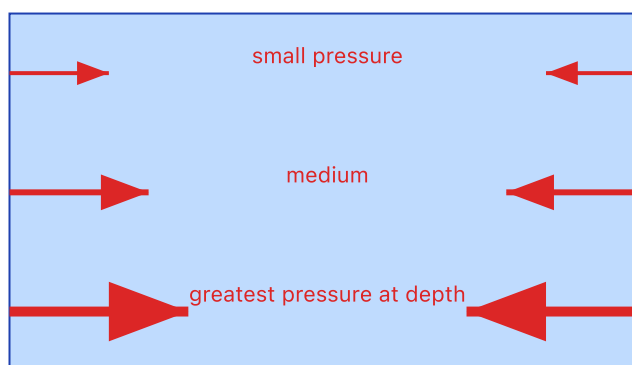
$$F = 250 \times 0.10 = \mathbf{25 \text{ N}}; E = \frac{1}{2} \times 250 \times 0.01 = \mathbf{1.25 \text{ J}}.$$

15.2 Pressure in Fluids

$$p = F / A \quad (\text{pascal} = \text{N/m}^2)$$

$$p = \rho \times g \times h \quad (\text{pressure at depth } h \text{ in a fluid})$$

Pressure Increases with Depth



Worked Example: Water density $1\,000\text{ kg/m}^3$, $g = 10$. Pressure at 15 m depth = $1\,000 \times 10 \times 15 = \mathbf{150\,000\text{ Pa}}$ ($= 1.5 \times 10^5\text{ Pa}$).

15.3 Upthrust and Floating

Upthrust arises because pressure on the *bottom* of a submerged object is greater than on the *top*. An object floats when upthrust = weight.

A steel ship floats because the hull shape displaces enough water that the average density of the ship is less than water.

15.4 Atmospheric Pressure

Air has weight. Atmospheric pressure at sea level $\approx 10^5\text{ Pa}$. Pressure **decreases** with altitude because there is less air above.

15.5 Hydraulic Systems

Pressure is transmitted equally in an incompressible fluid. A small force on a small piston produces a larger force on a bigger piston (same pressure $\times$ greater area). Used in car brakes and lifts.

Key Points Summary

- Hooke's law: $F = kx$ up to limit of proportionality; elastic PE = $\frac{1}{2}kx^2$.
- Pressure $p = F/A$ ($\text{Pa} = \text{N/m}^2$); in fluids $p = \rho gh$.
- Upthrust arises from pressure difference; floats when upthrust = weight.
- Atmospheric pressure $\approx 10^5 \text{ Pa}$, decreases with altitude.
- Hydraulics multiply forces using equal pressure across areas.